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CONTINUOUS INTERNAL ASSESSMENT – 1 Question Bank

23CSC02 – Machine Learning Techniques

PART -A

1. Find the Eigen value and Eigen vector for the below matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$
2. A Spam detection dataset contains 1000 emails, where 300 are spam. Compute the prior probability of spam and not spam. If an algorithm randomly selects an email. What is the probability it is spam?
3. Describe overfitting? How does cross-validation help to reduce it?
4. Define Bayes' theorem.
5. Compare and contrast generative and discriminative models
6. Compute the Projection of the A over B, A=[3,14] and B=[1,0]
7. Discuss gradient descent optimization.
8. Illustrate K-Nearest Neighbour algorithm. Mention one limitation.
9. Illustrate the role of kernel functions in Support Vector Machines?
10. Define supervised learning.
11. List any two applications of Machine Learning.
12. Differentiate classification and regression.
13. Given the vectors x = (2, −1, 3) and y = (1, 4, −2), Find the dot product.
14. Define a generative model.
15. Explain the least square method.
16. Describe the distance metric used in KNN.
17. Analyze the role of hyperparameters in Machine Learning.
18. Gmail automatically classifies emails as spam or not spam. Identify the ML application.
19. A real-estate company predicts house prices using size and location. Which model is suitable?
20. What do you understand by a random variable in probability and ML?

PART -B

1. Explain Classification using Logistic Regression and Support Vector Machines.
2. Explain any three types of discrete probability distribution function.
3. A medical test is used to detect a particular disease. The following information is known:

    a. The probability that a person has the disease is 0.05.
    b. The probability that the test correctly detects the disease (True Positive rate) is 0.90.
    c. The probability that the test incorrectly detects the disease in a healthy person (False Positive rate) is 0.10.

    If a person tests positive, what is the probability that the person actually has the disease?
    Use Bayes' Theorem to compute the required probability and interpret the result.
4. Suppose we have the following dataset with one response variable y and two predictor variables X1and X2

y	X_1	X_2
50	1	4
60	2	5
65	3	6
80	4	8

Fit a multiple linear regression model of the form

$y=\beta_0+\beta_1X_1+\beta_2X_2$

using the Least Squares (Normal Equation) method.

Express the model in matrix form.

Compute. $\hat{\beta} = (X^T X)^{-1} X^T Y.$ Write the final regression equation.

5. Compare underfitting and overfitting with neat diagram.
6. Illustrate the concept of Gradient Linear Models (GLMs) and Support Vector Machines (SVMs) as classification techniques.
7. Explain the working principle of the K-Nearest Neighbour (KNN) algorithm. Discuss the advantages and limitations of KNN as an instance-based learning method.
8. Describe and justify the importance of Machine Learning applications in healthcare
9. Analyze the advantages, disadvantages and challenges in implementing Machine Learning models.
10. Given P(A) = 0.4, P(B|A) = 0.7, and P(B|Ā) = 0.2:

    (a) Find P(B).

    (b) Find P(A|B).
11. Apply Bayes' theorem to explain how email spam filtering works.
12. Analyze the problems of underfitting and overfitting and discuss methods to overcome them.
13. Explain instance-based learning and the K-Nearest Neighbour algorithm.
14. A health analytics company wants to predict disease risk using 100 patient features, but many features are irrelevant. Explain how Lasso Regression can be applied to this problem.